

Product Requirement Document (PRD)

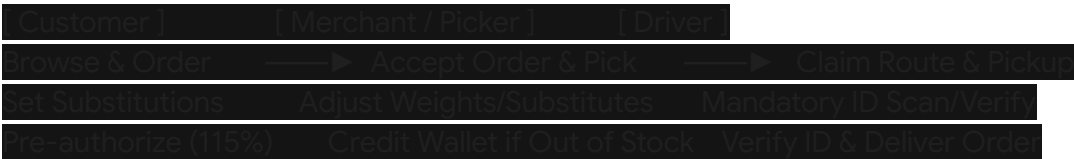
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Project Overview

- **Product Name:** On-Demand Grocery & Alcohol Delivery Marketplace
- **Core Architecture:** MedusaJS (Headless E-commerce Engine), Next.js (Storefront & Portals), PostgreSQL, Redis, Docker
- **Target Marketplace Model:** Multi-vendor (Customers, Merchants/Stores, Drivers/Pickers)
- **Primary Payment Gateway:** Yoco (via Yoco Hosted Checkout)

1. Executive Summary & Core Workflows

The platform enables customers to order groceries, produce, and alcohol from local merchants with options for immediate dispatch or scheduled time slots. It manages variable-weight items, product substitutions via a customer store credit/wallet, regulatory compliance, dynamic order picking, and mandatory driver ID verification on **all** drop-offs for security.



2. User Roles & System Permissions

Role	Access Level	Primary Objectives
Customer	Mobile Web / Web App	Browse catalogs, place orders, set substitution rules, manage in-app wallet balance, track deliveries in real time.
Merchant Admin / Picker	Web Dashboard	Manage product catalog, receive orders, pick items, adjust variable weights, trigger wallet credits for

		out-of-stock items.
Delivery Driver	Mobile Web Portal	Receive available routes, navigate to store/customer, complete mandatory ID verification at drop-off for all orders, mark fulfillment status.
Super Admin	Medusa Admin	Platform governance, toggle system features (Curfew Restrictions, On-Demand Fulfillment), commission management, payout processing, audit logs.

3. Functional Requirements

3.1 Drop-off ID & Delivery Verification

- **Mandatory Drop-off ID Verification:** To protect against fraud, carding, and unauthorized deliveries, **all deliveries (both Groceries and Alcohol)** require the driver to perform a mandatory ID verification (manual check or scanning) before unlocking delivery completion in the driver app.

3.2 Operating Hours & Curfew Restrictions

- **Curfew & Trading Hours Enforcement:** Restricts order placement for alcohol or overall store orders based on legally permitted or store-defined trading hours.
- **Admin Toggle Switch:** Super Admins must have an **Enable / Disable** toggle in the admin panel for Operating Hours & Curfew Restrictions.

3.3 Substitution & In-App Wallet Engine

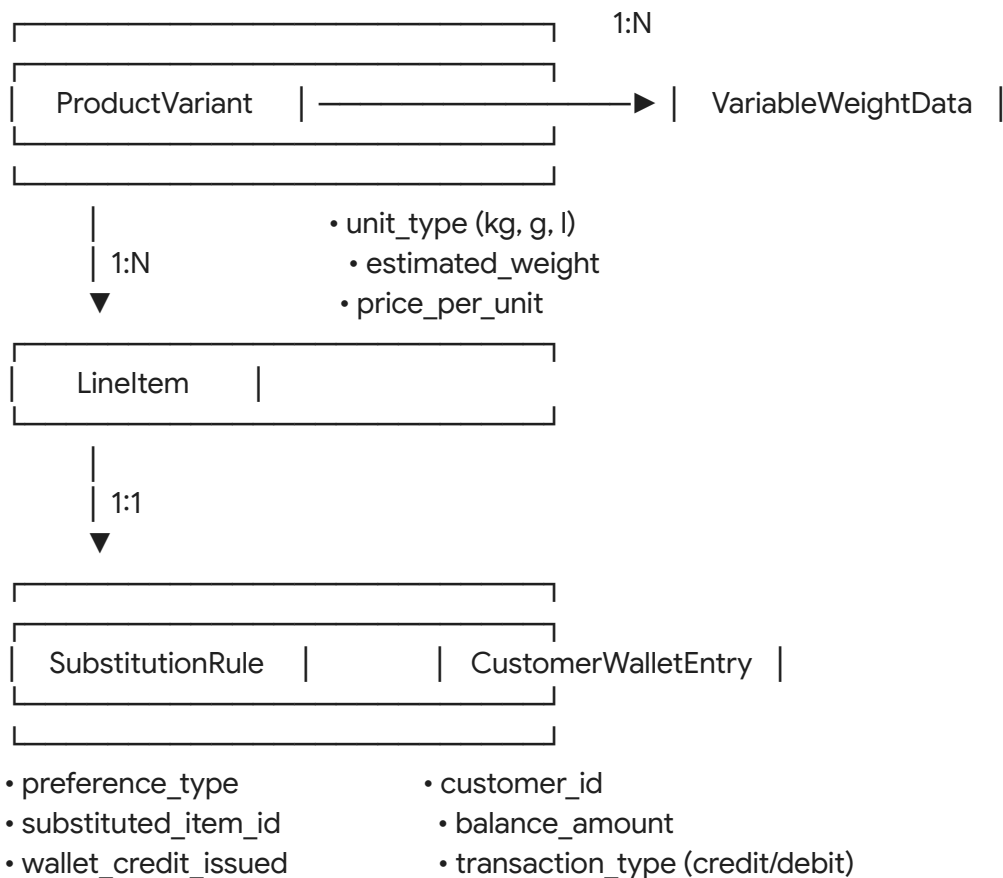
- **Pre-Authorization (115% Cap):** Payments are pre-authorized up to 115% of cart value via Yoco Hosted Checkout to accommodate weight variations or higher-cost item substitutions.
- **In-App Store Credit / Wallet System (No Automatic Card Refunds):**
 - To mitigate carding, refund fraud, and chargeback risks in South Africa, out-of-stock or unfulfilled items under the "Do Not Substitute" preference **do not issue automatic card refunds**.
 - Instead, unspent pre-authorized funds or missing item values are credited directly to the customer's **In-App Wallet / Credit Balance**, which can be applied toward future orders.
- **Substitution Rules:** Customers set item-level instructions during checkout:

- *Best Match* (Picker selects closest equivalent)
- *Contact Me* (In-app notification or SMS prompt)
- *Do Not Substitute* (Item value automatically credited to customer's wallet)

3.4 Delivery Slot & On-Demand Dispatch Control

- **Scheduled Time Slots:** Support for capacity-managed delivery windows (e.g., 2-hour delivery windows).
- **Immediate On-Demand Toggle:** Super Admins can **Enable / Disable** immediate on-demand fulfillment system-wide or per zone. When disabled, customers can only select scheduled delivery slots.

4. MedusaJS Custom Data Schema & Entities



Key Custom Modules Needed

1. **WalletModule:** Manages customer store credit balances, ledger entries, and wallet checkout deductions.
2. **ComplianceModule:** Handles mandatory drop-off ID verification records and admin

curfew toggles.

3. **FulfillmentToggleModule:** Controls immediate vs. scheduled delivery window availability based on admin settings.

5. Payment Gateway Integration & Security

- **Yoco Hosted Checkout:** Offloads card data collection, PCI-DSS compliance, and 3D Secure (3DS) authentication entirely to Yoco's secure hosted payment page.
- **Fraud Prevention Mitigation:** Eliminating direct card refunds in favor of wallet credits removes chargeback/carding exploits common with hijacked payment credentials.

6. System Architecture & Docker Setup

